

PayNear

MERN Full-Stack Project Proposal

Revised August 2026 - Owner Submission and Admin Publication Workflow

1. Project Summary

PayNear is a location-based web application that helps consumers discover nearby establishments based on accepted payment methods. Business owners can submit their store information and storefront image, while authorized PayNear administrators verify the information before it becomes publicly searchable.

2. Problem Statement

Consumers often learn only at checkout that a store does not accept their preferred payment method. Directory data can also be incomplete or outdated. PayNear addresses both problems through owner-submitted information and a controlled administrator verification process.

3. Proposed Solution

The public application provides verified search results, map-based discovery, payment filters, and store details. A separate owner area accepts private submissions. A protected administrator dashboard reviews each submission and controls whether the listing is published, returned for changes, rejected, or deactivated.

4. Project Objectives

- Help users find nearby verified establishments that accept a selected payment method.
- Give legitimate business owners a structured way to submit and maintain store information.
- Give administrators controlled moderation tools without requiring direct database access.
- Prevent pending, rejected, or inactive listings from appearing in public search results.

5. Target Users

- Consumers, travelers, and commuters who need payment-compatible establishments nearby.
- Business owners who submit store details, location, accepted payment methods, and an image.
- PayNear administrators, initially the project developers, who verify and publish listings.

6. MVP Features

Priority reflects the revised owner-to-admin publication workflow.

High Priority

- Role-based account access for consumers, business owners, and administrators.
- Consent-based location detection with manual coordinates as a fallback.
- Nearby search by radius, category, payment method, open status, and rating.
- Private owner submission of store name, category, address, coordinates, payment methods, and storefront image.
- Protected administrator queue with verify and publish, request changes, reject, edit, and deactivate actions.
- Public APIs that return only listings whose status is verified and active.
- Persistent MongoDB storage for accounts, moderation audit fields, images, and messages.

Medium Priority

- Favorites and a preferred payment method for authenticated consumers.
- Reviews and ratings that users can edit or remove only for their own account.
- Stable public links to verified establishment details.
- Real-time Socket.IO chat between a consumer and the verified establishment owner.

7. Out of Scope for the Initial MVP

- Payment processing or storage of card, bank, or e-wallet credentials.
- Automatic document authenticity decisions; administrators make the final verification decision.
- Turn-by-turn navigation, live traffic routing, and native Android or iOS applications.
- Automatic publication of any owner-submitted listing without administrator approval.

Scope clarification

Business-owner self-registration and store submission are now in scope. Automated verification remains out of scope: every listing must be reviewed by an authorized PayNear administrator.

8. High-Priority User Stories and Acceptance Criteria

US-01: Account Access

User story: As a consumer or business owner, I want to register and sign in so that I can securely use role-specific features.

- Registration validates required fields and prevents duplicate email addresses.
- Public registration can create consumer or owner accounts, but never an administrator.
- Protected routes reject missing, invalid, or expired sessions.

US-02: Location Permission

User story: As a consumer, I want to control access to my location so that nearby results are useful without sacrificing consent.

- The browser asks permission before location detection begins.
- Approved access provides usable coordinates for the current search.
- Denied or unavailable access preserves manual search and provides a helpful message.

US-03: Search and Filter

User story: As a consumer, I want to filter nearby establishments by payment method and radius so that I only see relevant options.

- Search supports category, payment method, radius, open status, and rating filters.
- Results within a location-based search fall inside the selected radius.
- Loading, empty, and error states are visible and understandable.

8. High-Priority User Stories (Continued)

US-04: Best-Match Results

User story: As a consumer, I want suitable establishments ranked for me so that I can choose quickly.

- Each result shows name, distance, address, status, image, and accepted payment methods.
- Only verified and active listings can appear as public results.
- Default ordering prioritizes distance and relevance.

US-05: Owner Store Submission

User story: As a business owner, I want to submit my store details and image so that PayNear administrators can review my business for publication.

- The owner provides name, category, complete address, coordinates, payment methods, and a valid image.
- A new submission begins as pending and is not publicly discoverable.
- The owner can view only their own private submissions and review notes.
- Sensitive edits to a verified listing return it to pending review.

US-06: Owner Image and Status Management

User story: As a business owner, I want to replace my storefront image and update operating status so that my listing stays useful.

- Image uploads accept JPG, PNG, or WebP files no larger than 3 MB.
- A replacement image requires another administrator review before publication.
- A verified owner can update open or closed status without changing administrator-only fields.

9. Administration and Supporting User Stories

US-07: Admin Verification and Publication

User story: As an administrator, I want to review owner submissions so that only trustworthy listings become public.

- Only authorized administrators can load the moderation queue or perform review actions.
- The administrator can verify and publish, request changes, reject, edit, or deactivate a listing.
- Verification requires a storefront image and records reviewer, review time, notes, and publication time.
- Review decisions notify the submitting owner.

US-08: Favorites and Payment Preference

User story: As a consumer, I want to save places and a preferred payment method so that repeat searches are faster.

- Favorites contain no duplicates and persist across authenticated sessions.
- The saved payment preference can be changed from the account experience.

US-09: Reviews and Shareable Details

User story: As a consumer, I want to review and share a verified establishment so that other people can make informed choices.

- A user can modify only their own review, and aggregate rating updates are displayed.
- A public link opens verified establishment details without exposing private account or user-location data.

10. Definition of MVP Complete

- All high-priority acceptance criteria pass functional tests.
- The consumer, owner, and administrator journeys work on mobile and desktop layouts.
- Role-based authorization protects ownership and administrator-only actions on the server.
- The deployed API connects securely to MongoDB and does not use the in-memory demo store.
- Uploaded listing images survive service restarts and redeployments.
- Pending, rejected, changes-requested, and inactive listings cannot appear in public results.

11. Technical Approach

Frontend	React and Vite provide responsive consumer discovery, owner submission, and protected administrator moderation screens.
Backend	Node.js and Express provide REST endpoints, validation, JWT authentication, ownership checks, moderation rules, and Socket.IO chat.
Database	MongoDB and Mongoose persist users, establishments, GeoJSON coordinates, moderation audit fields, images, messages, and notifications.
Deployment	Vercel hosts the React client. Render hosts the Express and Socket.IO service. MongoDB Atlas supplies production persistence.

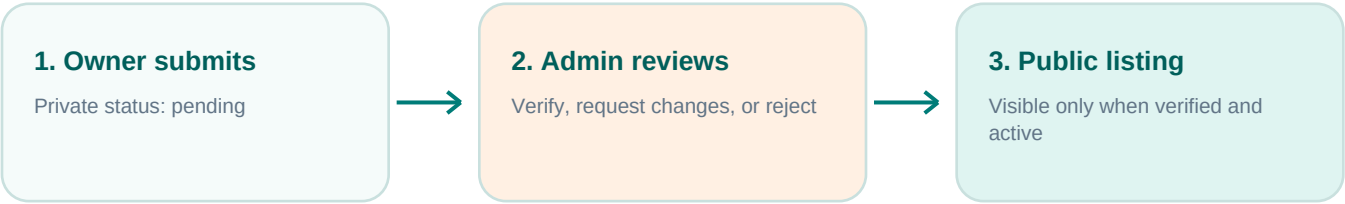
12. Quality and Security Requirements

- Server-side validation, hashed passwords, least-privilege authorization, and no public admin registration.
- Consent-based location use, sanitized inputs, image type and size limits, and protected moderation endpoints.
- Indexed geospatial queries, explicit loading and error states, and automated regression tests for publication rules.

13. Development Plan and Publication Workflow

- Phase 1 - Foundation: schemas, environment configuration, API conventions, and production database.
- Phase 2 - Access: registration, sign-in, session handling, owner authorization, and seeded administrator account.
- Phase 3 - Owner submission: store details, coordinates, payment methods, image persistence, and private listing status.
- Phase 4 - Administration: moderation queue, audit fields, review notes, verification, rejection, and publication.
- Phase 5 - Consumer discovery: verified-only search, map results, favorites, preferences, reviews, and sharing.
- Phase 6 - Release: responsive testing, security checks, production configuration, and deployment verification.

Listing State Lifecycle



14. Success Measures

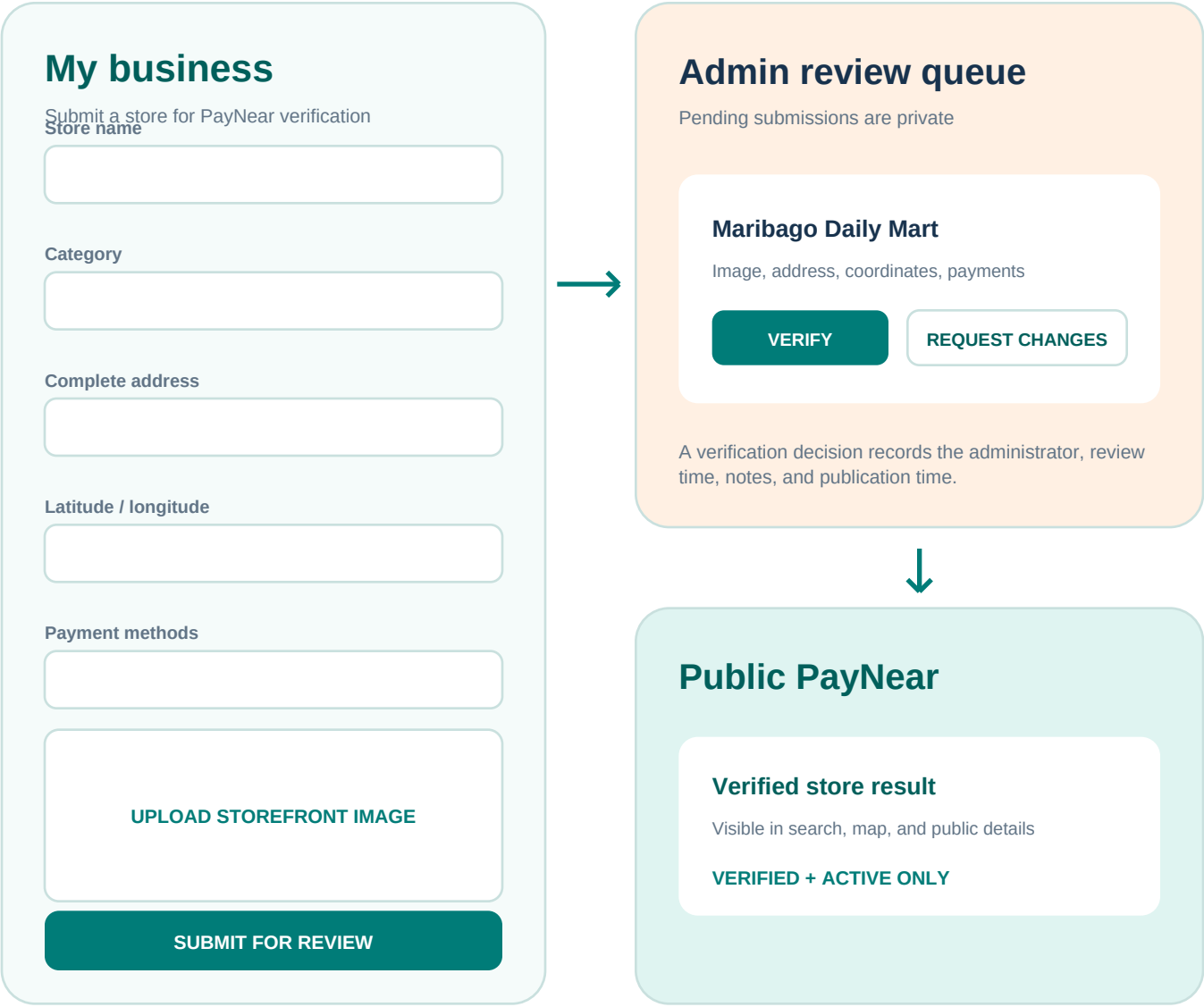
- Only verified and active establishments appear in public search and map results.
- Owners can submit and track stores without seeing or changing another owner's data.
- Administrators can moderate listings without direct database access.
- Publication decisions and owner-sensitive edits are traceable through audit fields and automated tests.

15. Requested Decision

Approve PayNear as a MERN full-stack project with consumer, business-owner, and administrator roles, using the owner submission and administrator verification workflow as a required part of the MVP.

16. Updated Wireframes - Owner Submission

WF-OWNER-01: private store submission and administrator handoff



Interaction sequence

1. Owner registers and submits complete store evidence. 2. The listing remains private while administrators review it. 3. Verification publishes it to consumers. 4. Rejected or changes-requested submissions remain private and return to the owner with notes.

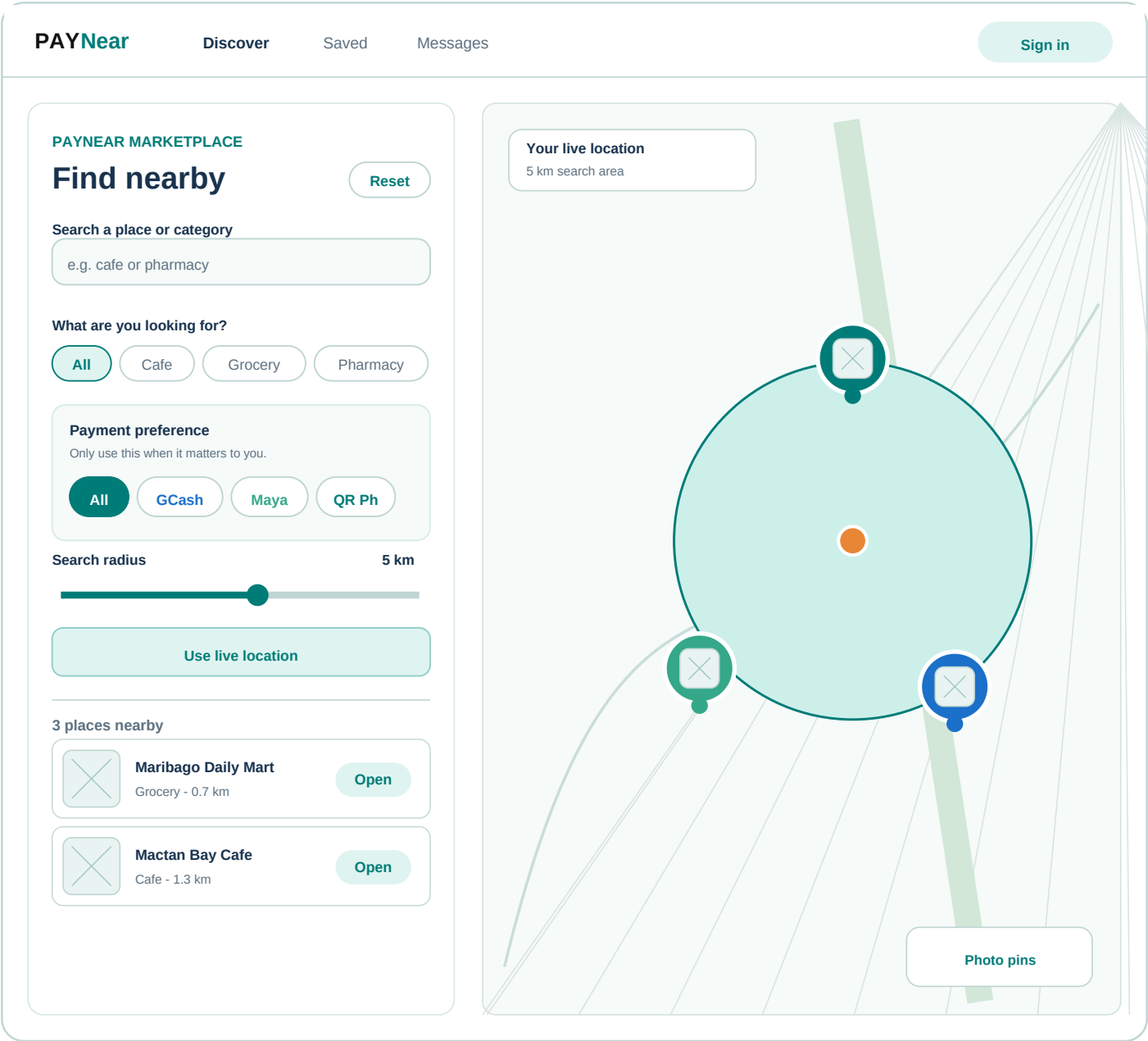
15. Updated Wireframes

Updated July 30, 2026. This section replaces the wireframe note on page 7.

WF-01

Discover nearby - primary desktop screen

One sidebar keeps search, payment availability, radius, and matching establishments in a single flow.



Wireframe notes

The map uses the visitor location after permission, with a visible radius circle. Store images identify establishments on the map; payment badges communicate accepted methods only and do not process a payment.

15. Updated Wireframes (continued)

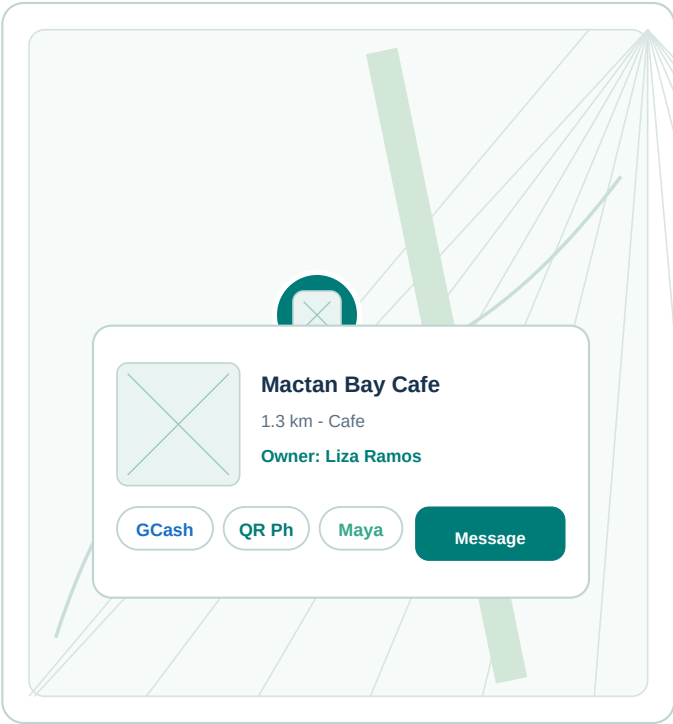
The visitor can inspect a photo pin, then begin a real-time conversation with the listed contact.

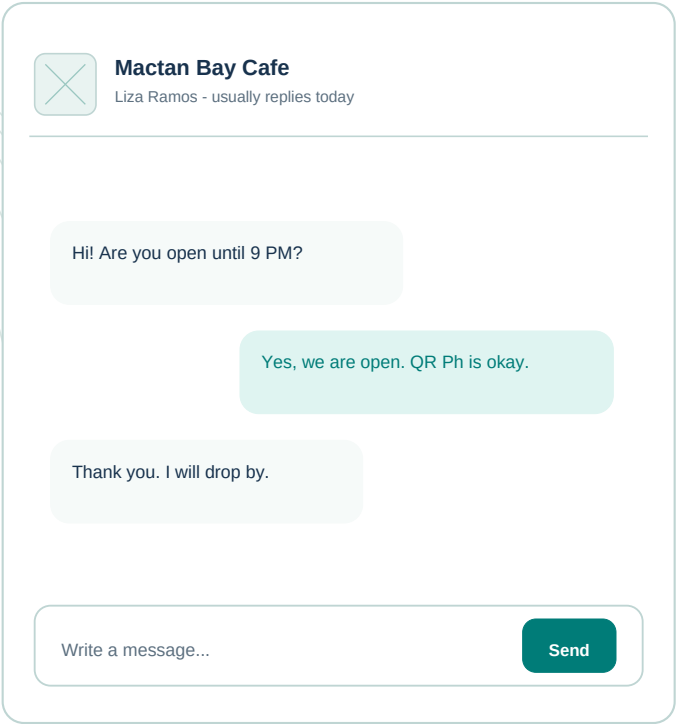
WF-02

Photo pin and establishment preview

WF-03

Real-time chat with owner





Tap a photo pin to view the listing, its owner, distance, accepted methods, and contact action.

Socket.IO keeps the conversation current for both visitor and establishment contact without reloads.

Interaction sequence



The preview does not imply payment processing. It simply helps visitors find an establishment whose listed payment options match their needs, then communicate directly with the establishment.

15. Updated Wireframes (continued)

THE LAST ROOM

Supporting screens keep listing data manageable for administrators and practical on a smaller mobile screen.

WF-04 Admin listing management

PayNear Admin

New listing

Establishment details

Name: Maribago Daily Mart

Category: Grocery

Address: Maribago, Lapu-Lapu City

Contact: Miguel Reyes

Listing photo

UPLOAD

Accepted payment methods

GCash

Maya

QR Ph

InstaPay

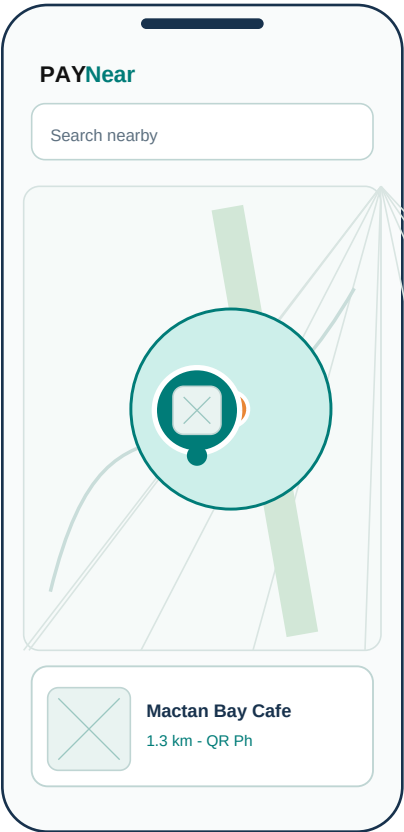
Save changes

Archive

Verify

Admin tools manage listing details, images, availability, and verification.

WF-05 Mobile discovery



On mobile, search and the nearby result stay within reach while the radius remains visible.

Implementation alignment

Images

Listing photos are uploaded and become map identifiers.

Payments

GCash, Maya, QR Ph, InstaPay, banks, cards, cash, and transfer can be listed.

Roles

Visitors discover and message; administrators create, update, archive, and verify listings.

End of updated wireframes